

Computational Solid Mechanics

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Overview

- Element formulations
- Constitutive laws
 - Plasticity
 - Composites
 - Thermo mechanical
- Contact mechanics
- Isogeometric analysis
- Cable-nets
- Sensitivity analysis
- Optimization



Solid Mechanics in Kratos

- Strategies

- Newton-Raphson
- Line search
- Arc length (force control)



Solid Mechanics in Kratos

- Strategies

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- Arc length

- Elements

- Small displacement elements (**solids** (2D, 3D, axysym), **beams** (2D and 3D Euler-Bernoulli, locking-free 2D straight and curved Timoshenko), 3D and 2D **trusses**, thin and thick **Shell, membranes**)
- Updated and Total Lagrangian (solids, 3D truss, 3D Euler-Bernoulli beams)
- Mixed $\mathbf{u-p}$ and $\epsilon_{vol}-\mathbf{u}$
- Bbar elements

- Incoming elements:

- Mixed $\mathbf{u}-\epsilon$ for solids, alleviate mesh dependency
- 3D locking free 2 noded and 3 noded straight/curved Timoshenko elements
- 2D plane strain Timoshenko beam
- 3D 3 noded Truss
- 3, 4, 6 and 8 noded direct integrated shells



- Strategies

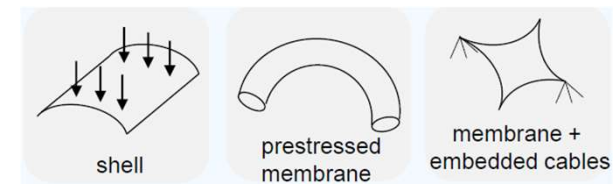
- Newton-Raphson
- Line search
- Arc length

- Elements

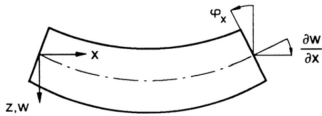
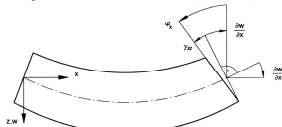
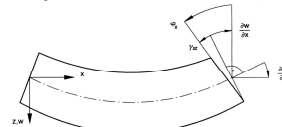
- Small displacement elements (solids (2D, 3D, axysym), beams (2D and 3D Euler-Bernoulli, locking-free 2D straight and curved Timoshenko), 3D and 2D trusses, thin and thick Shell, membranes)
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Shell Elements in Kratos

	shell_thin_element_tri	shell_thick_element_tri	shell_thin_element_quad	shell_thick_element_quad
Model	3-parameter (KL) 	5-parameter (RM) 	3-parameter (KL) 	5-parameter (RM) 
Membrane	Assumed Natural Deviatoric Strain (ANDES)	Pure displacement	ANDES	Enhanced Assumed Strain (EAS)
Bending	Discrete Kirchhoff Triangular (DKT) ²	Pure displacement	Discrete Kirchhoff Quadrilateral (DKQ)	Pure displacement
Shear	-	Discrete Shear Gap (DSG) + Stenberg stabilization	-	Mixed Interpolation Tensorial Components (MITC) + Stenberg stabilization
Drilling DOFs	ANDES	Artificial drilling stiffness	ANDES	Artificial drilling stiffness
Non-linear analysis	extend the capability of linear small strain elements with co-rotational transformation (element-independent CR (EICR) formulation)			

Courtesy of Ricky Aristio



Solid Mechanics in Kratos

- Strategies

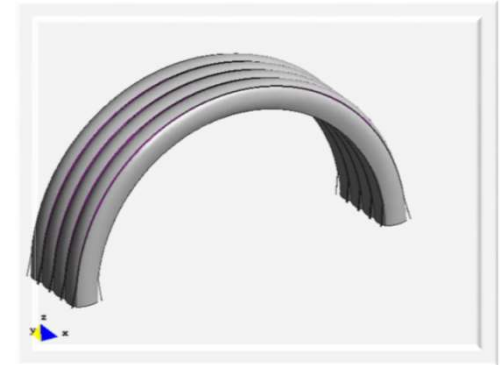
- Newton-Raphson
- Line search
- Arc length

- Elements

- Small displacement elements (axysymm, solid, beam, truss, Shell, etc)
- Updated and Total Lagrangian
- Mixed $u-p$ and $\epsilon_{vol}-u$
- Bbar elements

- Constitutive laws

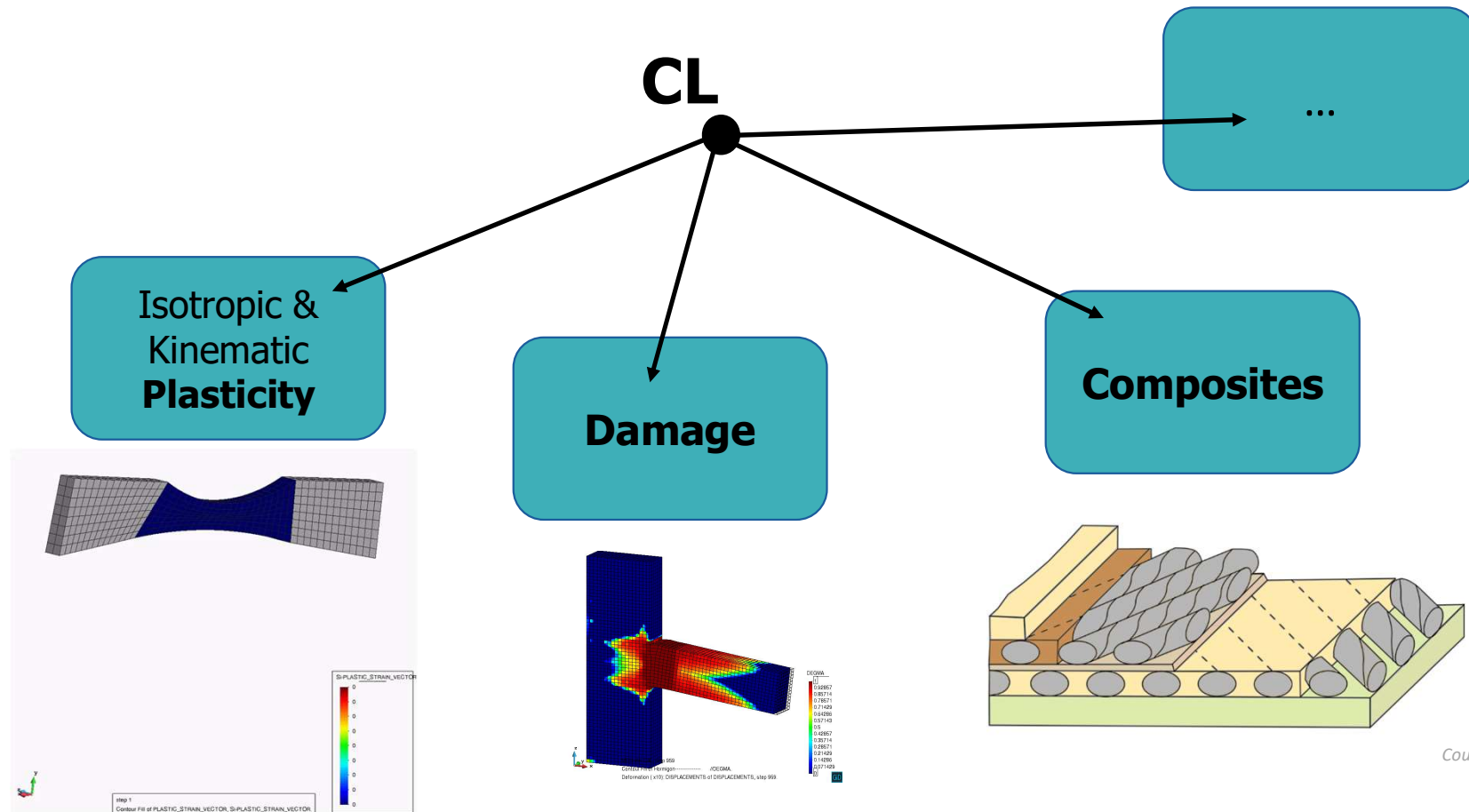
- Elasticity
- **Hyper-elasticity** (Neo-hookean, Kirchhoff...)
- **Viscoelasticity** (Kelvin and Maxwell)
- **Plasticity** (isotropic and/or kinematic; Non-associative; Finite strains $\log(\mathbf{E})$, many flow rules)
- Isotropic **Damage**
- **Delamination** in composites
- Directional d+d- damage
- Orthotropic damage
- Plastic-Damage
- **Composite** materials (Classical and SP-RoM)
- High cycle **fatigue**
- **Anisotropy**/Orthotropy



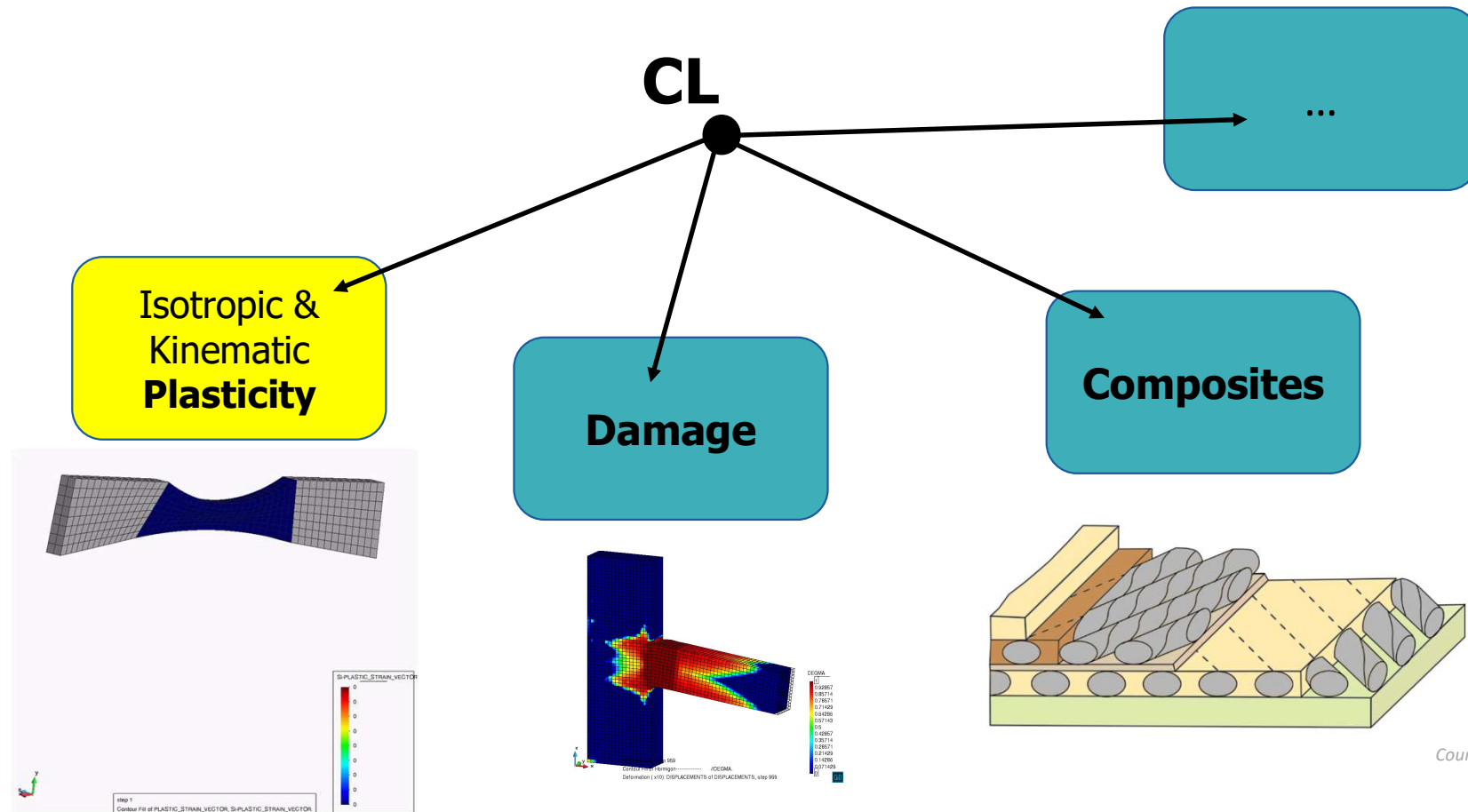
Courtesy of A. Cornejo



Modular implementation of advanced CL



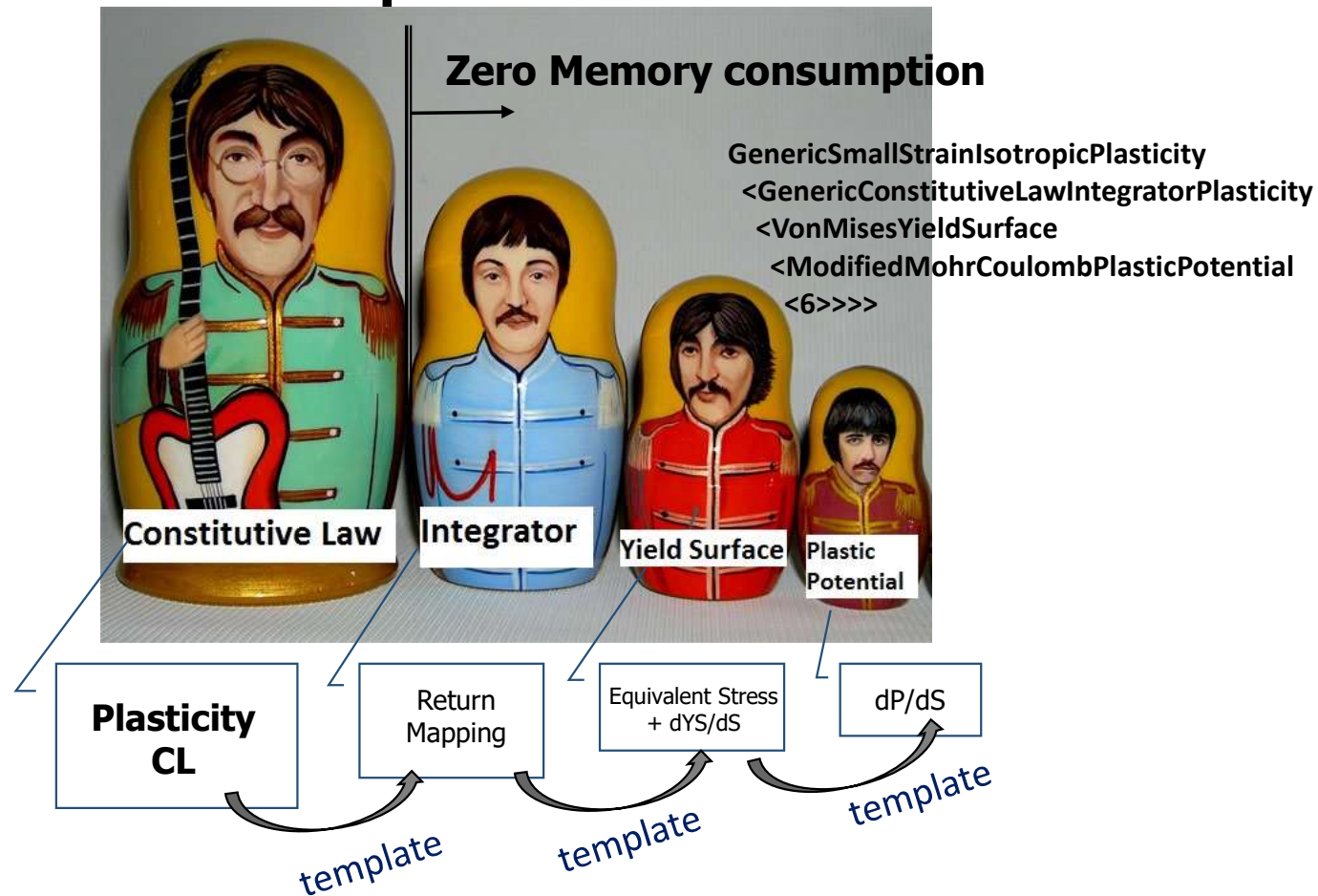
Modular implementation of advanced CL



Courtesy of A. Cornejo



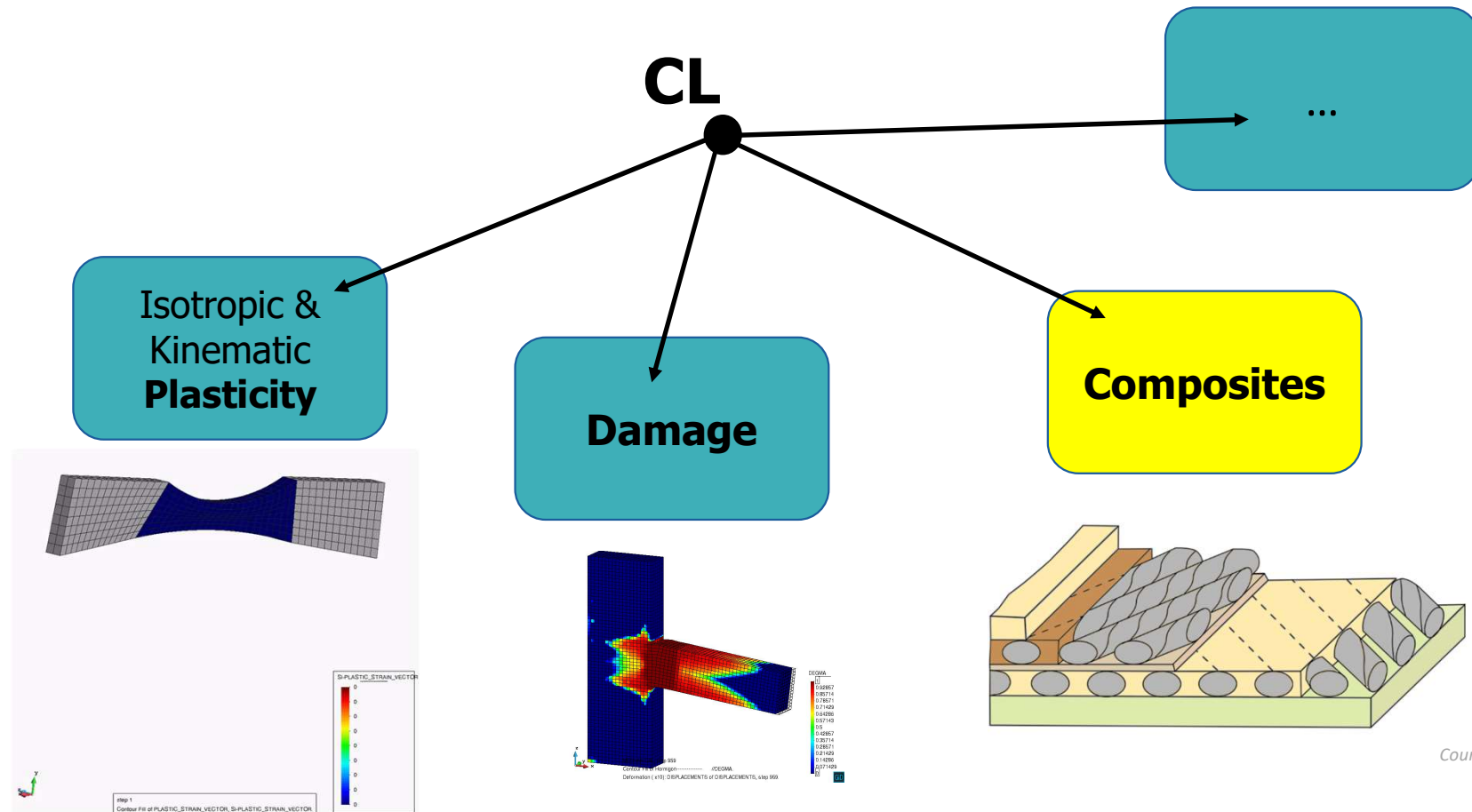
Modular implementation of advanced CL



Courtesy of A. Cornejo



Modular implementation of advanced CL



Courtesy of A. Cornejo

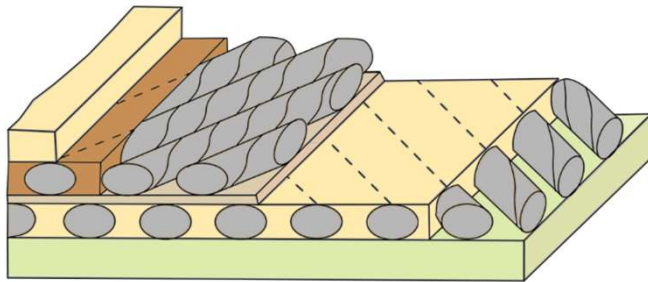


Properties and Subproperties

Rule of Mixtures

CL's inside another CL

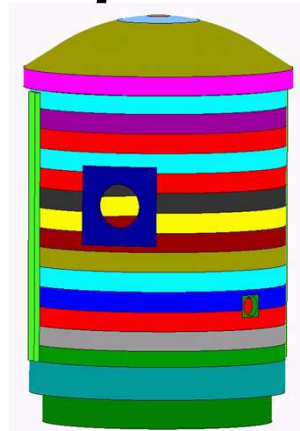
Local Axes, rotations



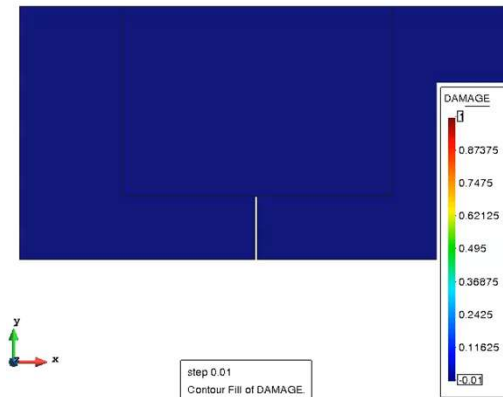
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      "combination_factors" : [0.5, 0.5]  
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        "POISSON_RATIO" : 0.35  
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      "POISSON_RATIO" : 0.2,  
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},  
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}
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Courtesy of A. Cornejo

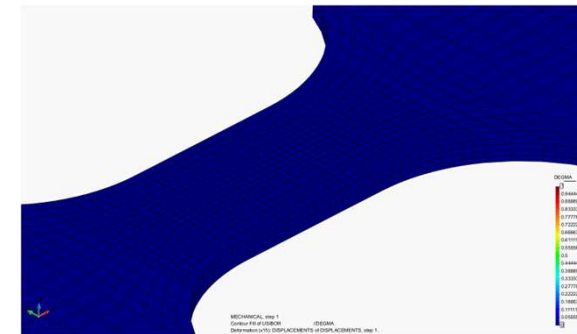
Examples



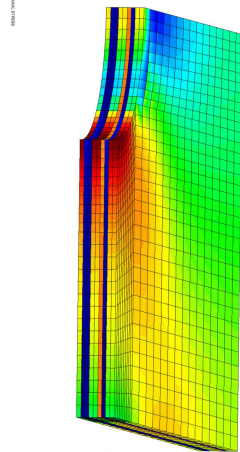
Pre-stressed concrete



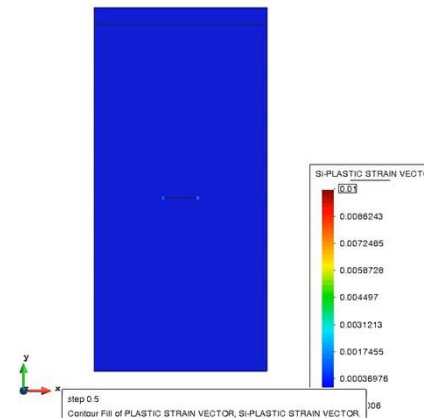
Fracture mechanics



High cycle fatigue of metals

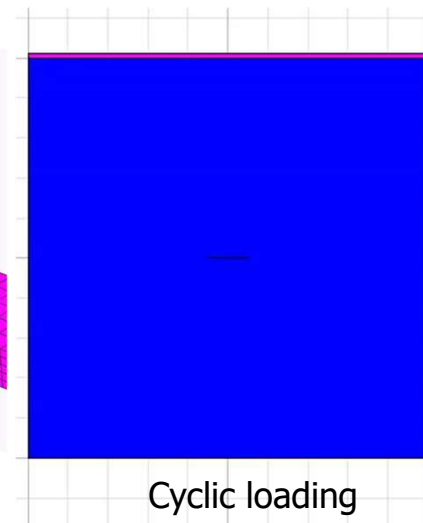
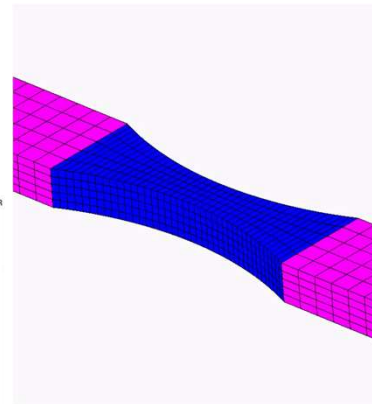


Composite materials



Plasticity

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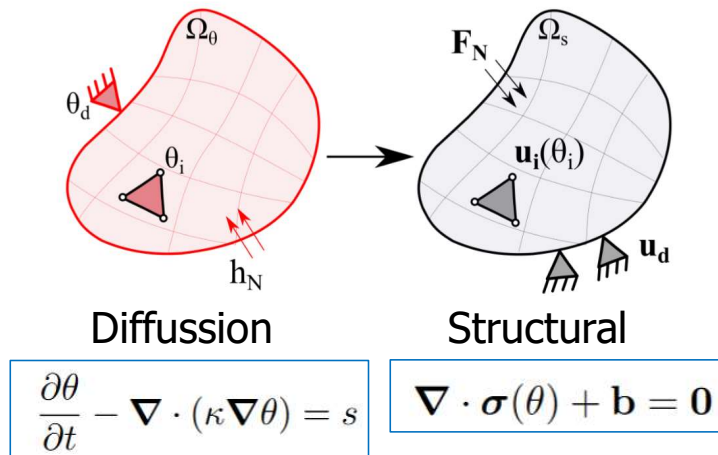


Cyclic loading

Courtesy of A. Cornejo



Thermo Mechanical

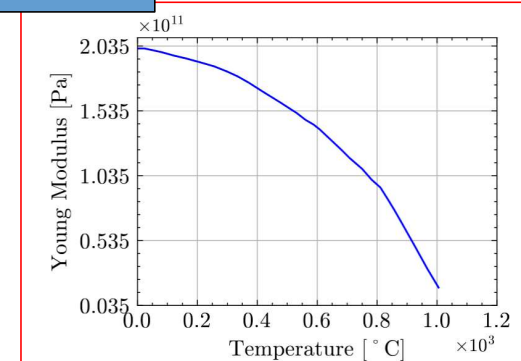
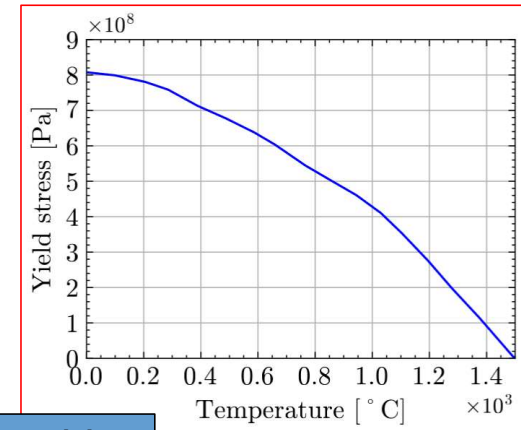


Temperature **affects material properties** (via Accessor)

Temperature dependent **strains** (volumetric)

Cracking due to Thermal gradients (damage, fatigue, etc.)

Input Tables



Courtesy of A. Cornejo



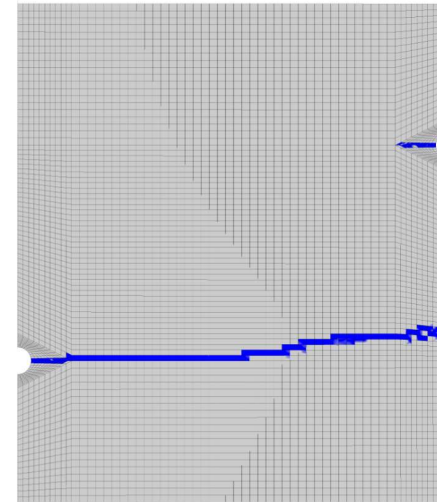
Thermo Mechanical



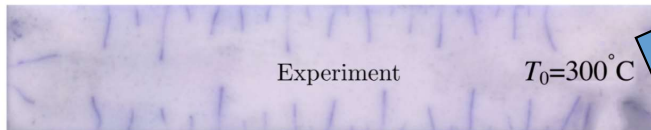
$$\theta(t) = 1200 \cdot \sin(\pi \cdot t/6) + 100 \text{ }^{\circ}\text{C}$$



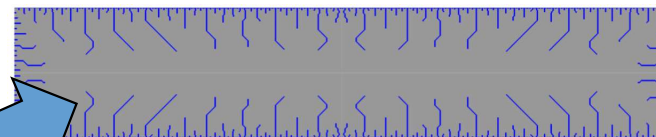
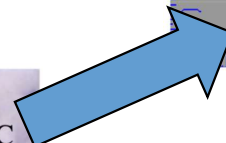
Thermally induced crack
due to fatigue



Quenching test of concrete



$T_0 = 300^{\circ}\text{C}$

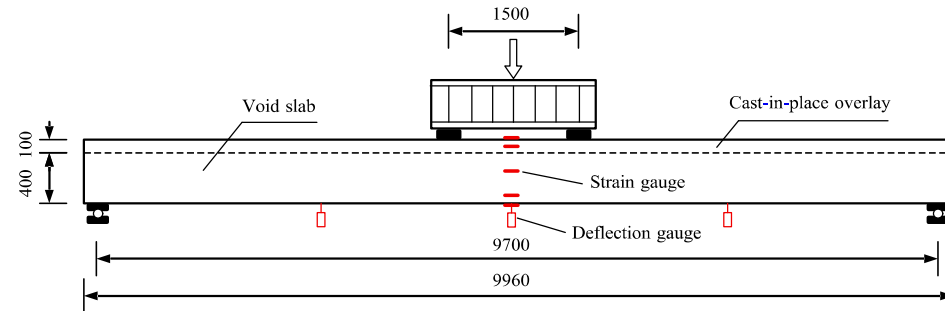
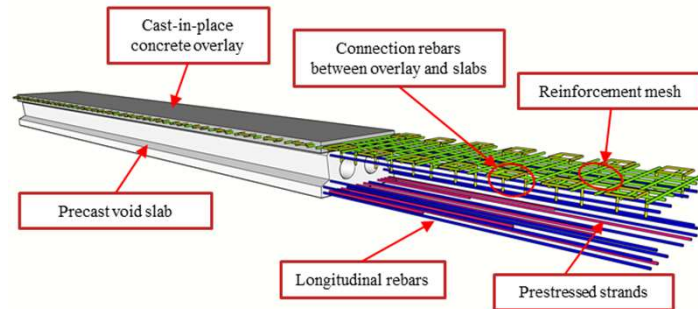


Thermal SHOCK cracking

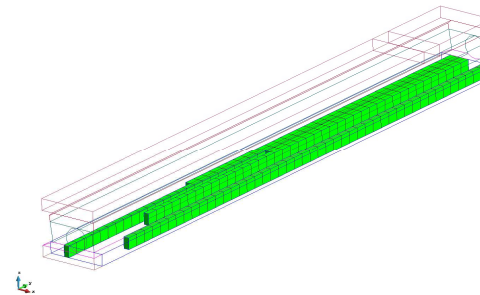
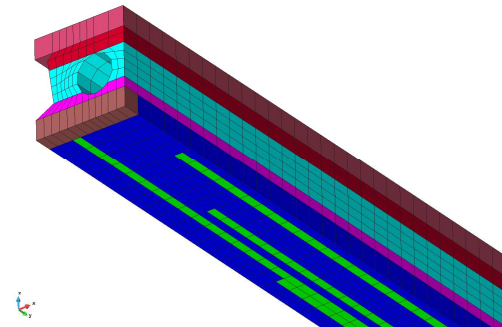
A hot sample baths in cold water...



Pre-stressed concrete simulations



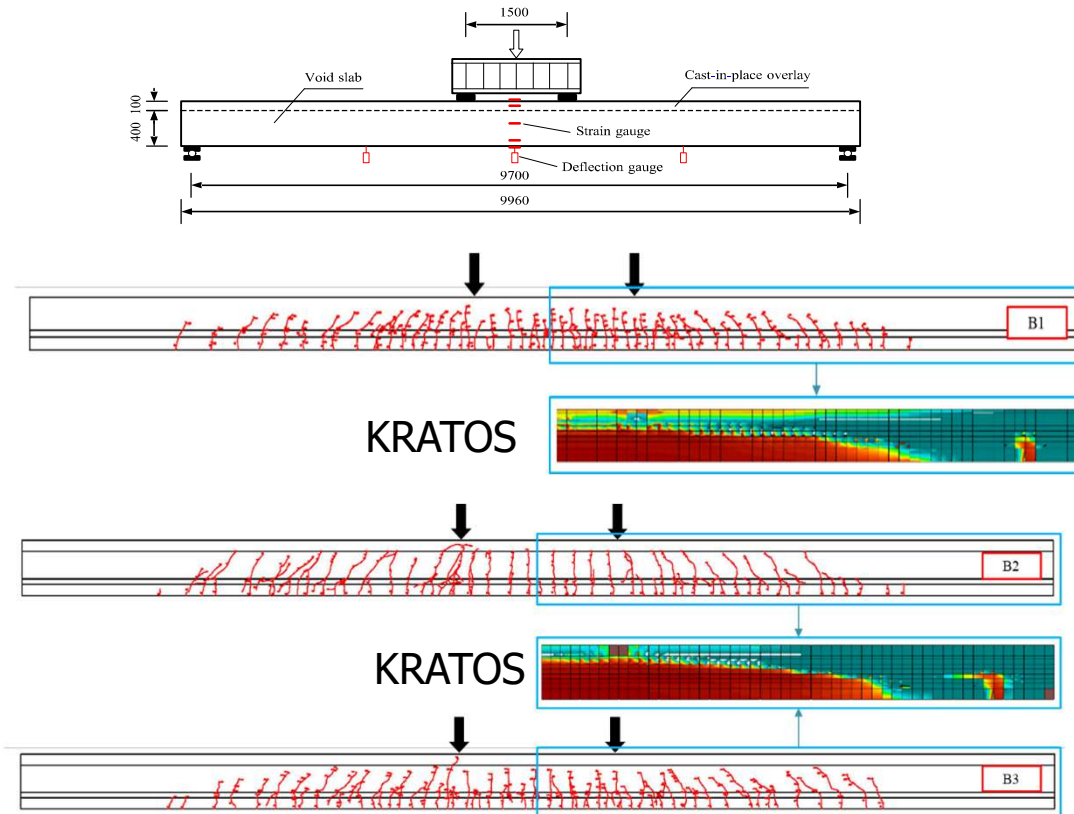
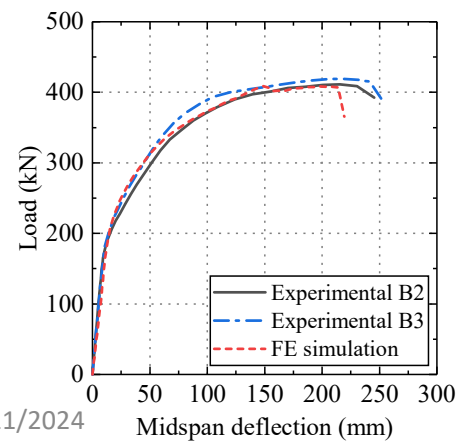
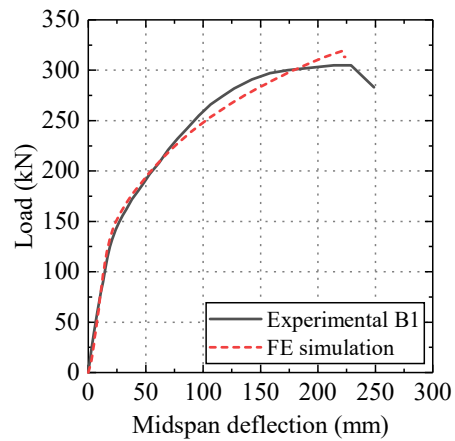
Concrete (Isotropic damage)		
Property	Value	Units
Young Modulus (E)	35230	MPa
Poisson ratio (ν)	0.2	-
Tension yield strength (f_t)	3.74	MPa
Compression yield strength (f_c)	44	MPa
Fracture energy (G_f)	150	J/m ²
Friction angle (ϕ)	32	deg
Steel rebars (Von Mises isotropic plasticity and mapping theory)		
Orthotropic Young Modulus [E_{xx} , E_{yy} , E_{zz}]	[200000, 36764, 36764]	MPa
Orthotropic Poisson ratios [ν_{xy} , ν_{yz} , ν_{xz}]	[0.3, 0.3, 0.3]	-
Orthotropic strengths [$f_{t,xx}$, $f_{t,yy}$, $f_{t,zz}$, $f_{t,xy}$, $f_{t,yz}$, $f_{t,xz}$]	[412, 71.65, 71.65, 71.65, 71.65, 71.65]	MPa
Ultimate stress f_u (isotropic space)	580	MPa
Pre-stressed steel (Von Mises isotropic plasticity and mapping theory)		
Orthotropic Young Modulus [E_{xx} , E_{yy} , E_{zz}]	[199259, 36764, 36764]	MPa
Orthotropic Poisson ratios [ν_{xy} , ν_{yz} , ν_{xz}]	[0.3, 0.3, 0.3]	-
Orthotropic strengths [$f_{t,xx}$, $f_{t,yy}$, $f_{t,zz}$, $f_{t,xy}$, $f_{t,yz}$, $f_{t,xz}$]	[1764, 306.8, 306.8, 306.8, 306.8, 306.8]	MPa
Ultimate stress f_u (isotropic space)	1931	MPa



Courtesy of A. Cornejo



Pre-stressed concrete simulations



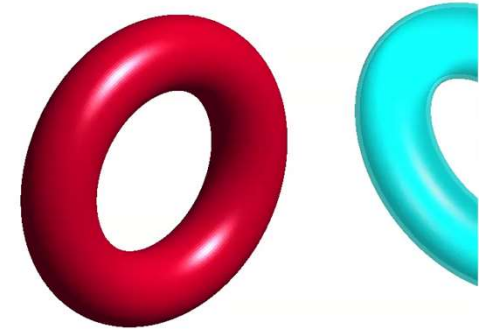
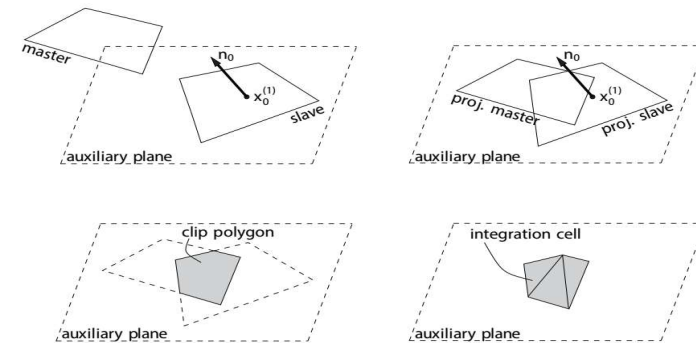
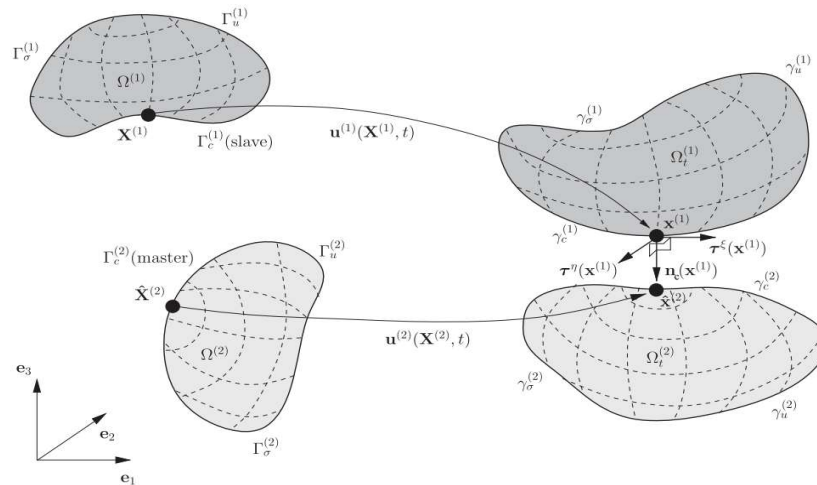
Courtesy of A. Cornejo



Computational Contact Mechanics

- **Methodologies to model implicit frictional contact**

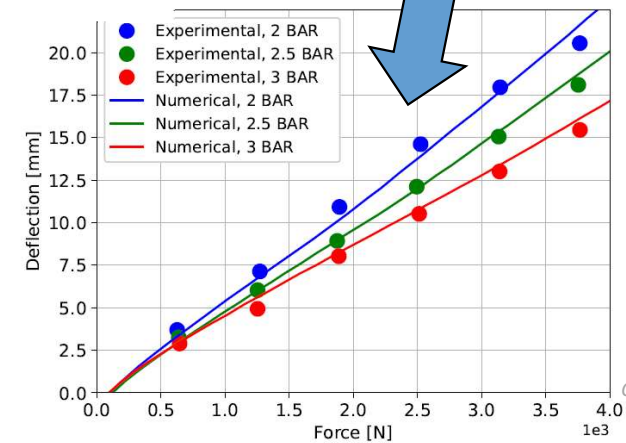
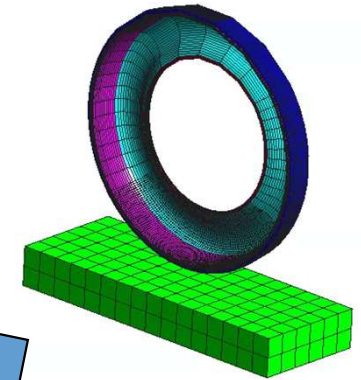
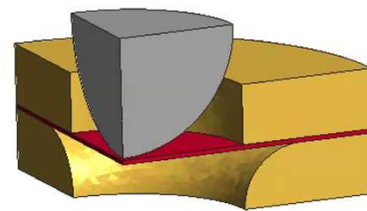
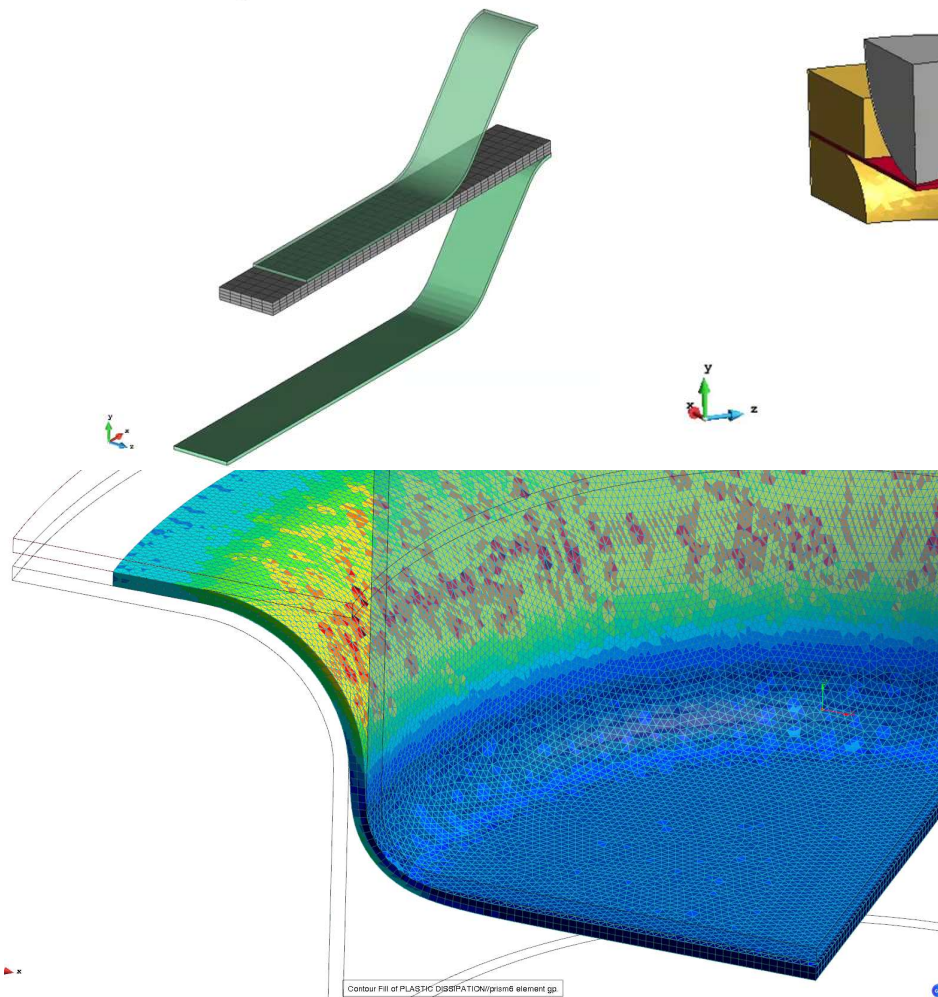
- Dual **Augmented Lagrangian** Method
- **Mortar** Method for estimating the gap
- **Implicit** frictionless/**frictional** contact



Courtesy of V. Mataix



Computational Contact Mechanics

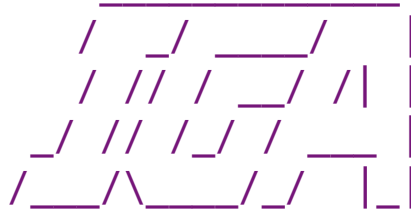


Courtesy of V. Mataix

Isogeometric Analysis



KRATOS



Application

CAD-Integrated Structure



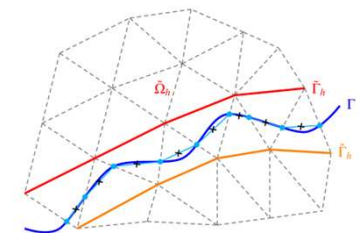
Courtesy of T. Teschemacher & M. Meßmer

Fluid-Structure Interaction



Courtesy of P. Bücher

Shifted boundary method



Courtesy of R. Zorilla

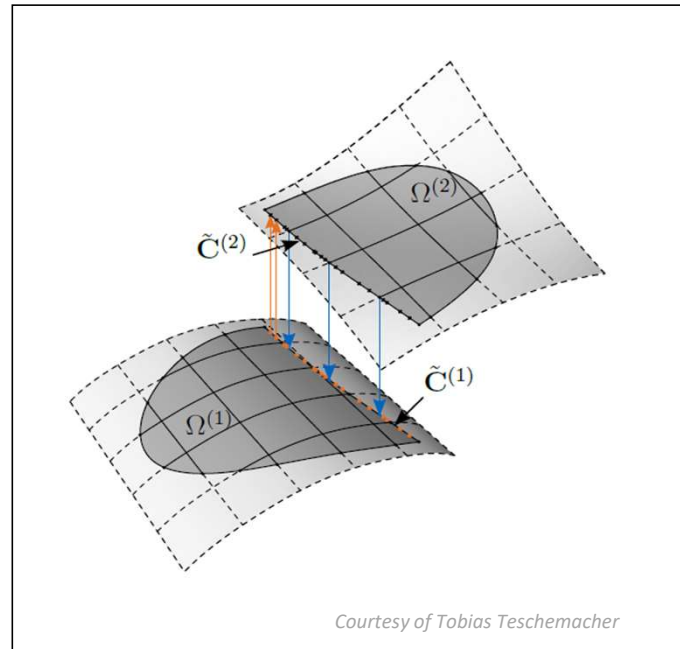
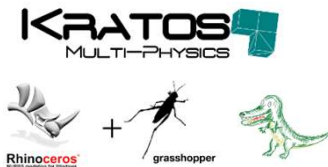


Isogeometric Analysis

Isogeometric
elements

- Kirchhoff-Love shells [Kiendl et al.]
- Reissner-Mindlin shells [Echter et al., ...]
- Membranes
- Convection Diffusion SBM [Antonelli et al.]
- ...

CAD-Integrated
workflow



Courtesy of Tobias Teschemacher

Trimming
(integration
method)

- Tesselation & Triangulation
- Moment-fitting and point elimination

Multi-patch
Coupling (weak imposition
of constraints)

- Penalty
- Lagrange Multipliers
- Nitsche-type
- ...

Courtesy of Ricky Aristio

6/11/2024 – 8/11/2024

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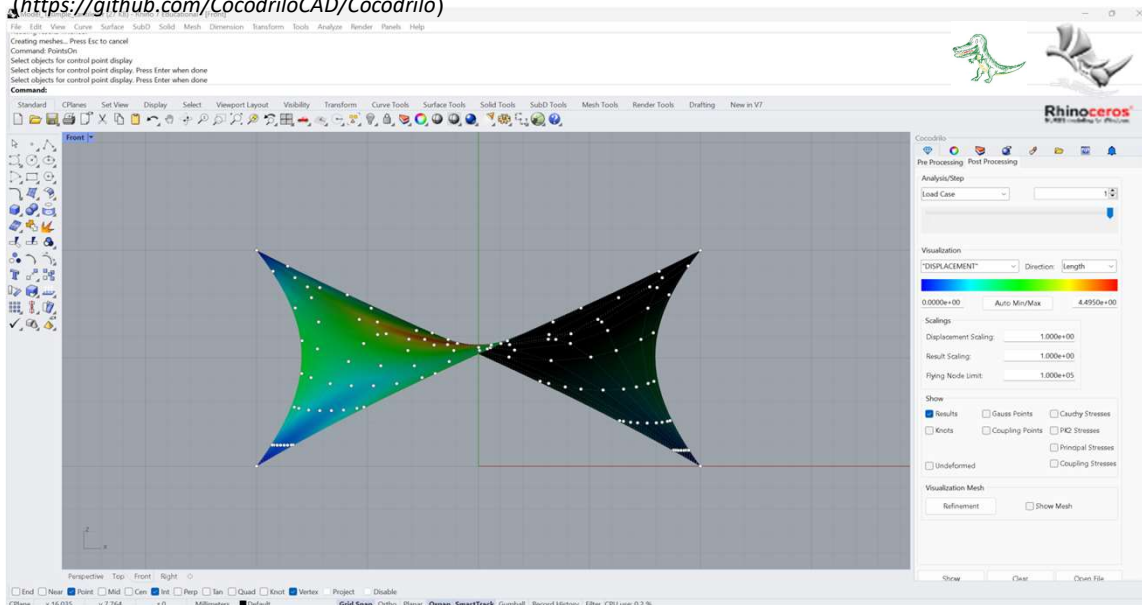


Isogeometric Analysis

- Design-through-analysis workflow

Postprocessing: Cocodrilo

(<https://github.com/CocodriloCAD/Cocodrilo>)

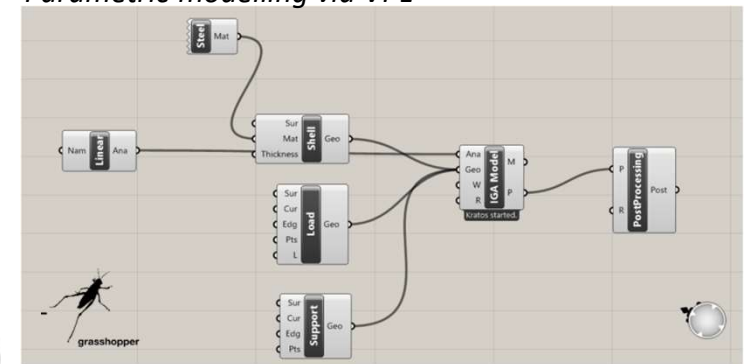


Courtesy of Ricky Aristio

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Preprocessing: Cocodrilo

Parametric modelling via VPL



```
D:\Cocodrilo\Cocodrilo\Cocodrilo\bin\FPS_2_elements_test_v9\python kratos_main_iga.py
Kratos
Multi-Physics 9.3.0-7f3c3cedf4-Release-x86_64
Compiled for Windows and Python3.10 with MSVC-1932
Compiled with threading support.
Maximum number of threads: 20
Importing KratosIgaApplication
Kratos
Initializing KratosIgaApplication...
Importing KratosStructuralMechanicsApplication
Kratos
StructuralMechanics
Initializing KratosStructuralMechanicsApplication...
[WARNING] DEPRECATION-Warning; MechanicalSolver: Using "block_builder", please move it to "builder_and_solver_settings"
as "use_block_builder"
[WARNING] DEPRECATION-Warning; MechanicalSolver: Using "line_search", please move it to "solving_strategy_settings" as "
type"
::[MechanicalSolver]:: : Construction finished
::[FormingMechanicalSolver]:: : Construction finished
```

Solver: Kratos Multiphysics

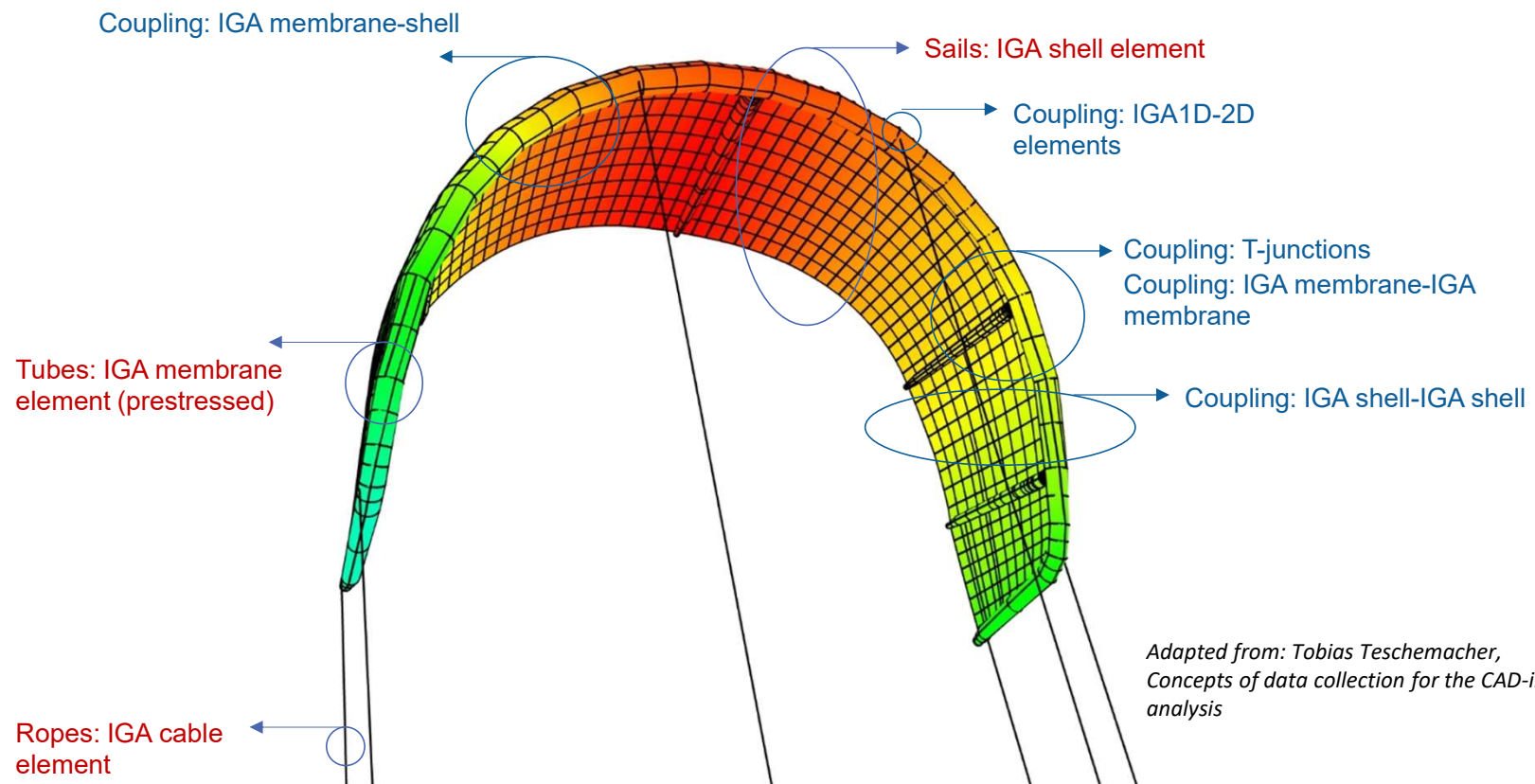
(<https://github.com/KratosMultiphysics/Kratos>)



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Isogeometric Analysis

- CAD-Integrated Structure: Kite example



Courtesy of Ricky Aristio

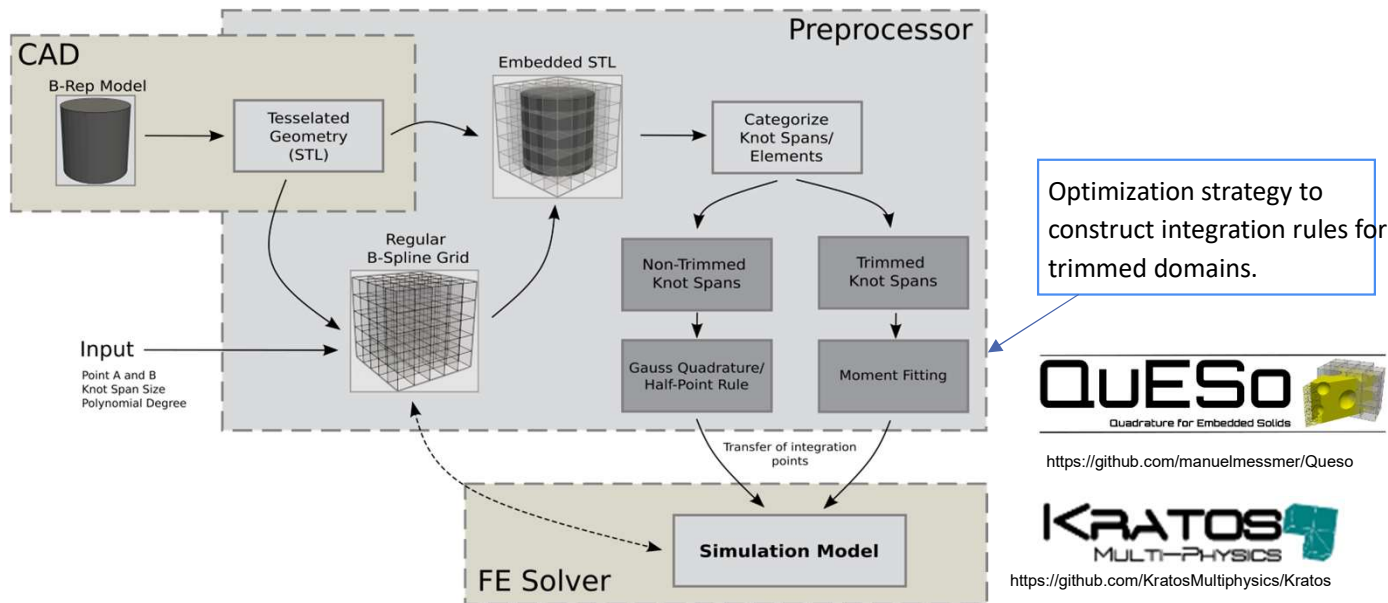
Adapted from: Tobias Teschemacher,
Concepts of data collection for the CAD-integrated isogeometric analysis



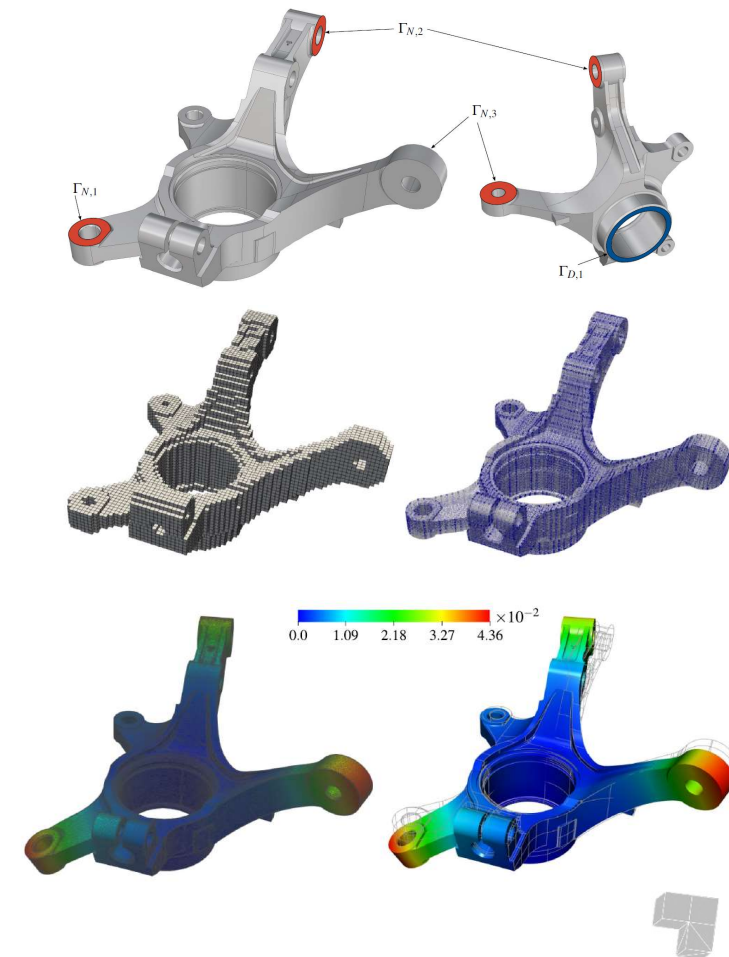
Isogeometric Analysis

- Workflow for embedded solids

Any CAD Software



Courtesy of Manuel Meßmer

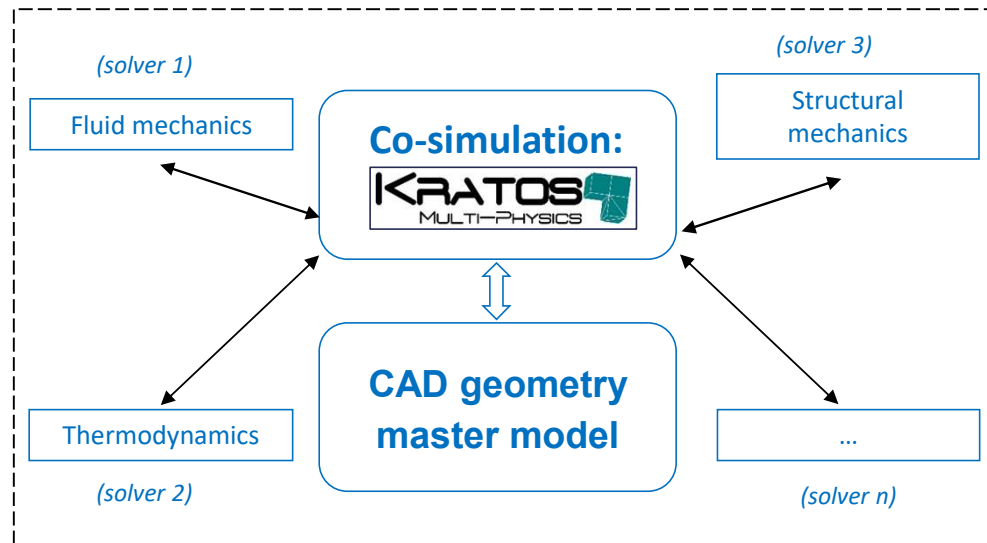


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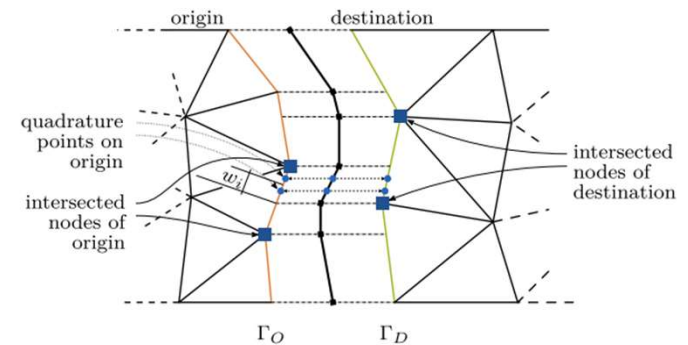
Isogeometric Analysis

- Fluid-Structure Interaction
 - via co-simulation



Mapping methods

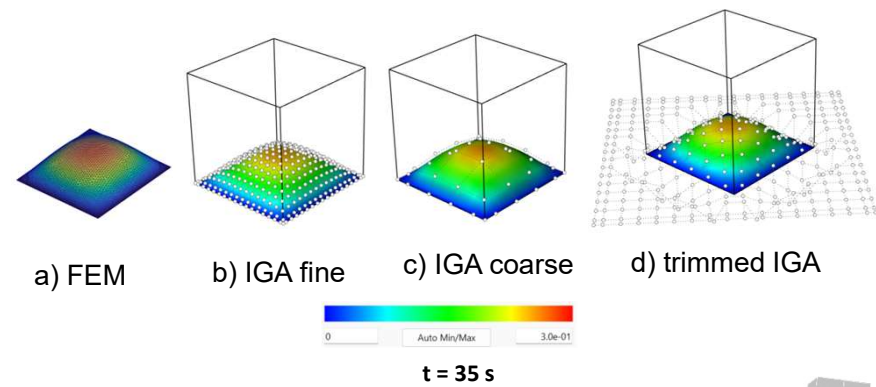
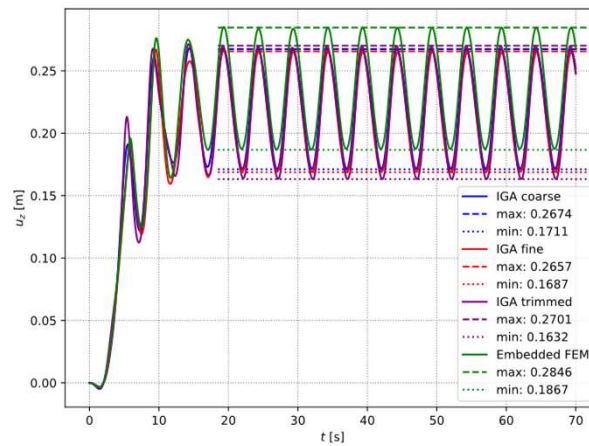
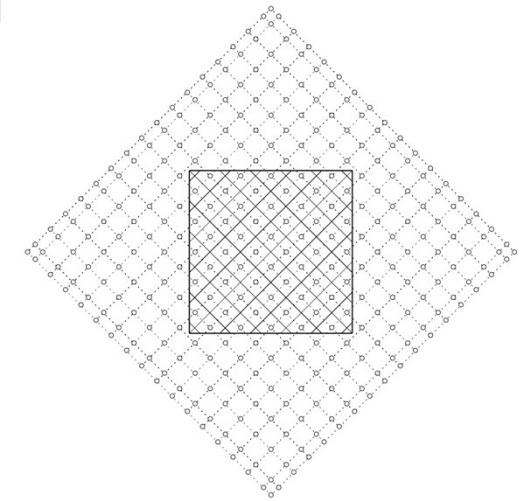
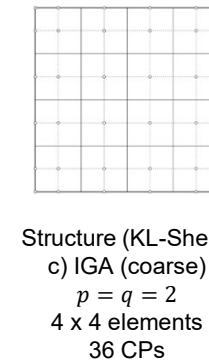
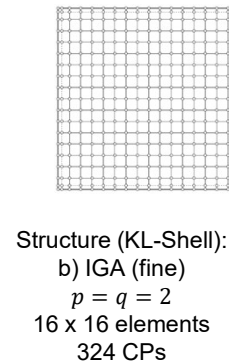
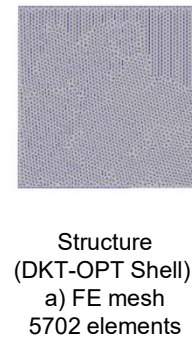
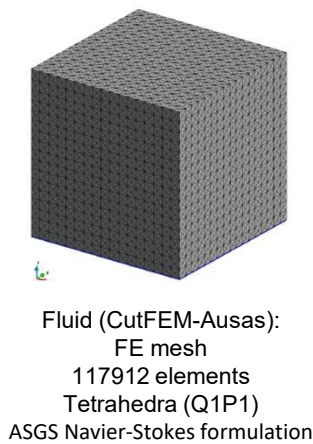
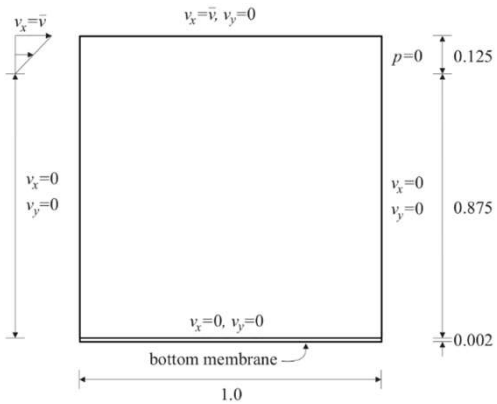
- Nearest Neighbour
- Nearest Element
- **Mortar-based mapping**
- ...



Courtesy of Wilson et al.

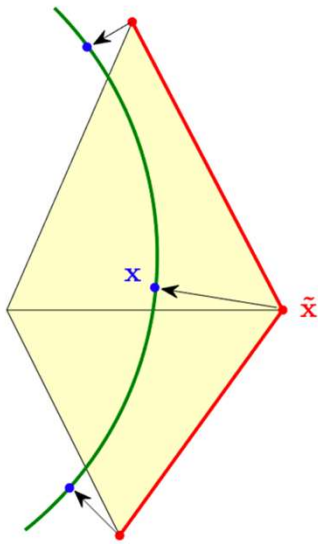


Isogeometric Analysis - Benchmark



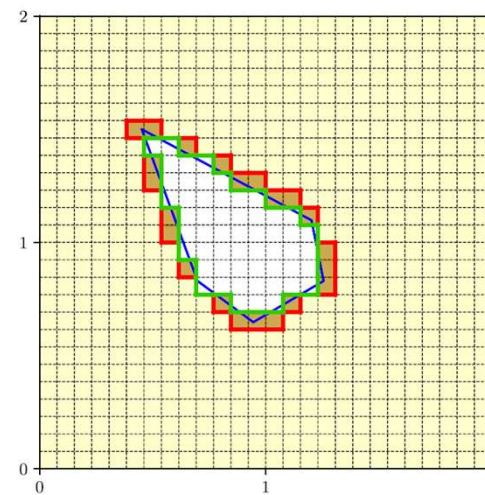
Isogeometric Analysis - SBM

— Skin Boundary Γ
— Shifted Boundary $\tilde{\Gamma}$

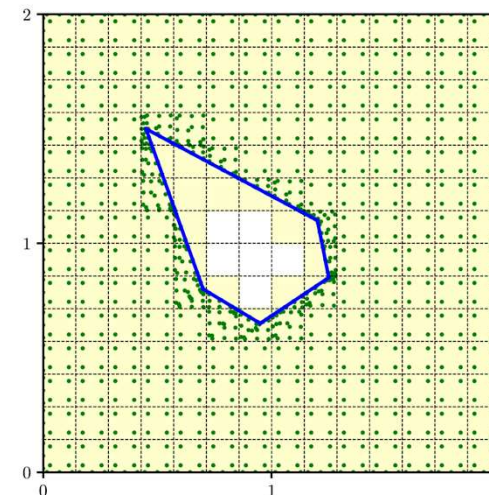


→ available for Poisson problem
(on progress: Navier Stokes, linear elasticity)

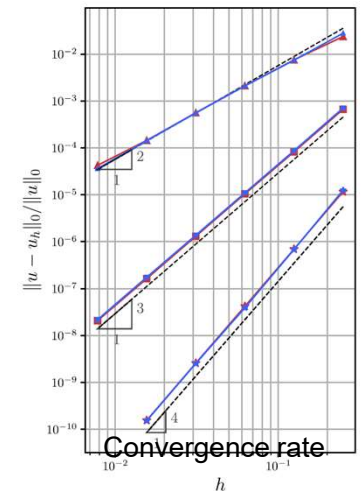
Courtesy of N. Antonelli



SBM set-up



IBRA set-up



Impose *modified* Dirichlet boundary conditions at the shifted boundary

- 2D Poisson problem:

$$-\Delta u = f \quad \text{in } \Omega,$$

$$u = u_D \quad \text{on } \Gamma_D,$$

$$\nabla u \cdot \mathbf{n} = t_N \quad \text{on } \Gamma_N,$$

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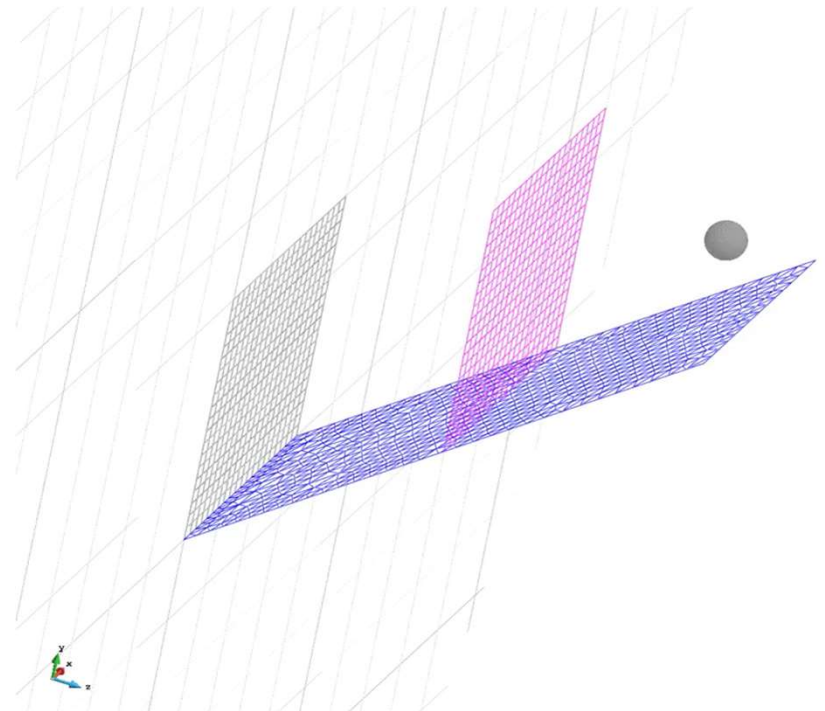
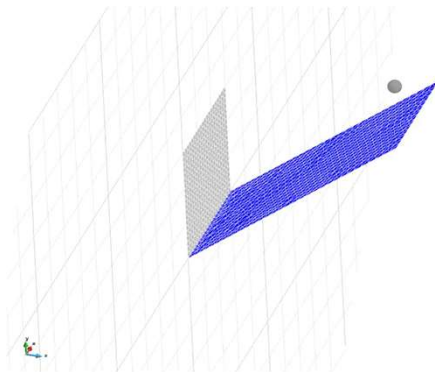
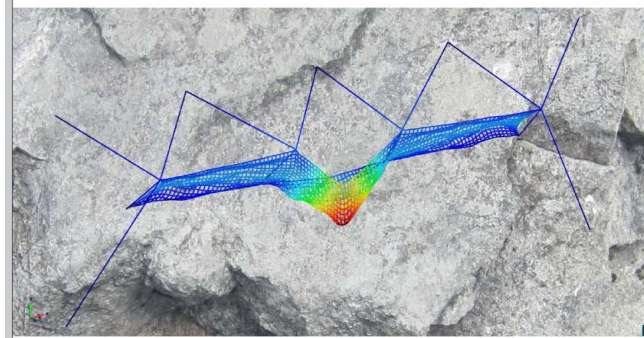
Taylor expansion:

$$u(\mathbf{x}) = u(\tilde{\mathbf{x}}) + \nabla u|_{\tilde{\mathbf{x}}} \cdot (\mathbf{x} - \tilde{\mathbf{x}}) + o(|(\mathbf{x} - \tilde{\mathbf{x}})|^2)$$

$$u(\tilde{\mathbf{x}}) \approx u(\mathbf{x}) - \nabla u|_{\tilde{\mathbf{x}}} \cdot (\mathbf{x} - \tilde{\mathbf{x}})$$



Cable-nets



Courtesy of Klaus Sautter

6/11/2024 – 8/11/2024

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Sensitivity Analysis

Ingredients for the optimization process :

→ Objective: $\min J(\mathbf{x}, p)$

Ex: strain difference minimization, by
adjusting “M” parameters p

→ Primal solution: $\mathbf{R} = \mathbf{K}\mathbf{x} - \mathbf{F} = \mathbf{0}$

Ex: Solve for “N” $\mathbf{x}$ displacements

→ Sensitivity analysis – variants:

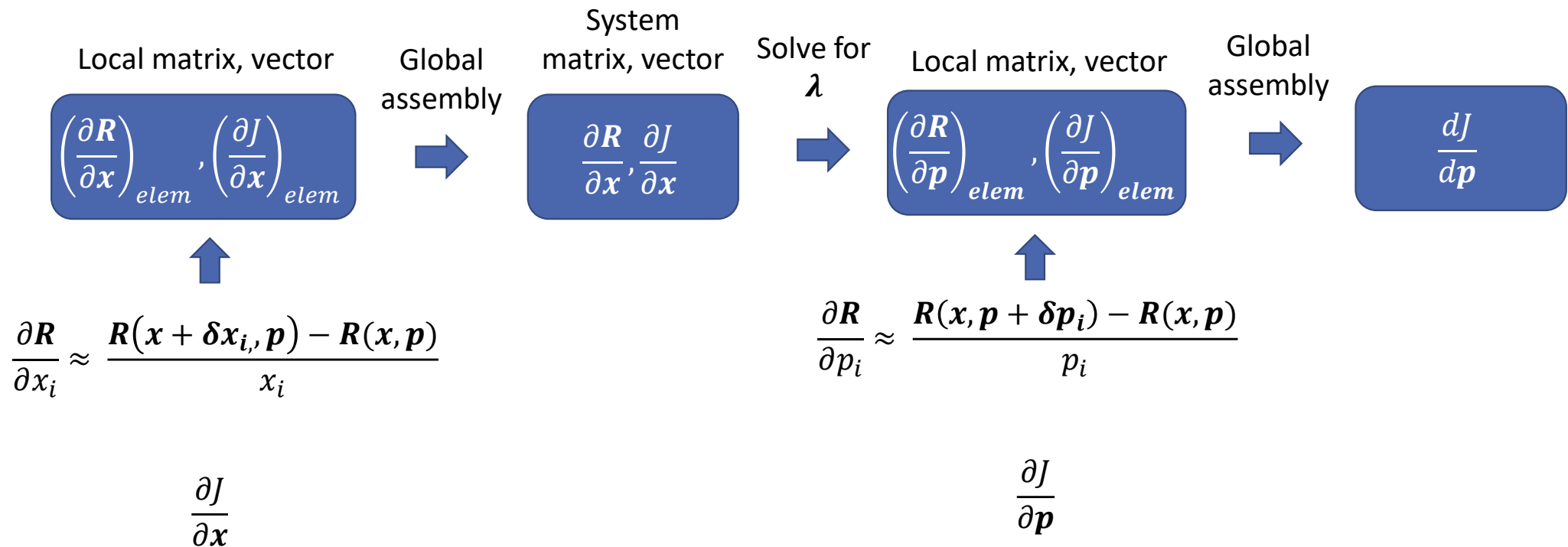
Direct approach: $\frac{dJ}{dp} = \frac{\partial J}{\partial p} + \left[\frac{\partial J}{\partial \mathbf{x}} \right]^T \frac{d\mathbf{x}}{dp} \rightarrow$ Requires (M+1) primal solves

Adjoint approach: $\frac{dJ}{dp} = \frac{\partial J}{\partial p} + \lambda^T \frac{\partial \mathbf{R}}{\partial p}$
 $\lambda^T \leftarrow \left[\frac{\partial \mathbf{R}}{\partial \mathbf{x}} \right]^T \lambda = - \left[\frac{\partial J}{\partial \mathbf{x}} \right] \rightarrow$ Requires only one adjoint solve

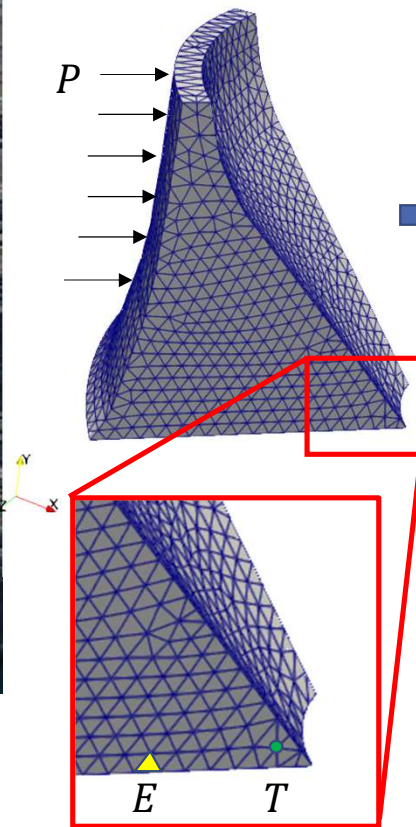
The cost for the evaluation of gradients is
independent of the number of variables used for p !



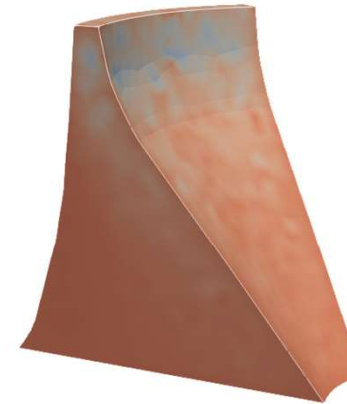
Sensitivity Analysis



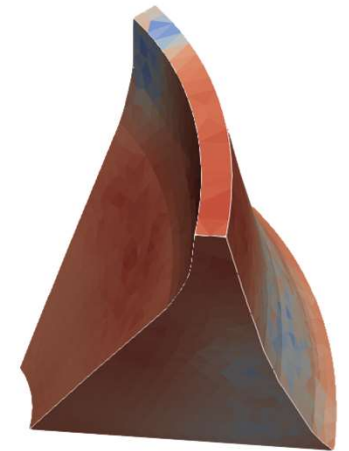
Sensitivity Analysis



w.r.t. P



w.r.t. T



w.r.t. E

Measurement residual sensitivity

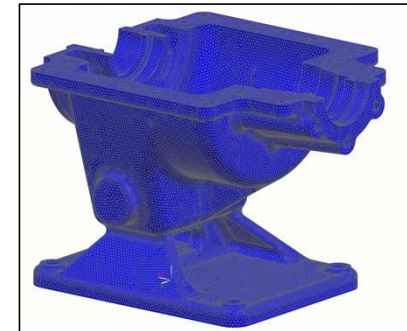
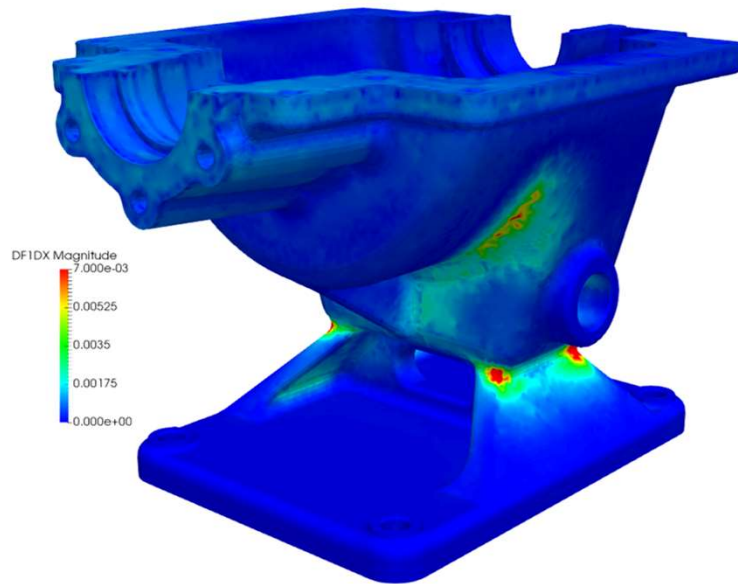
$$J = \frac{1}{2} \sum_{i=1}^M w_i (u_i^c - u_i^m)^2$$

Courtesy of Talhah Ansari



Sensitivity Analysis

- Sensitivity map for design guidance



- Eigenfrequency #1
- Identification of hot spots
- Added-value even without optimization!
- Complex geometries

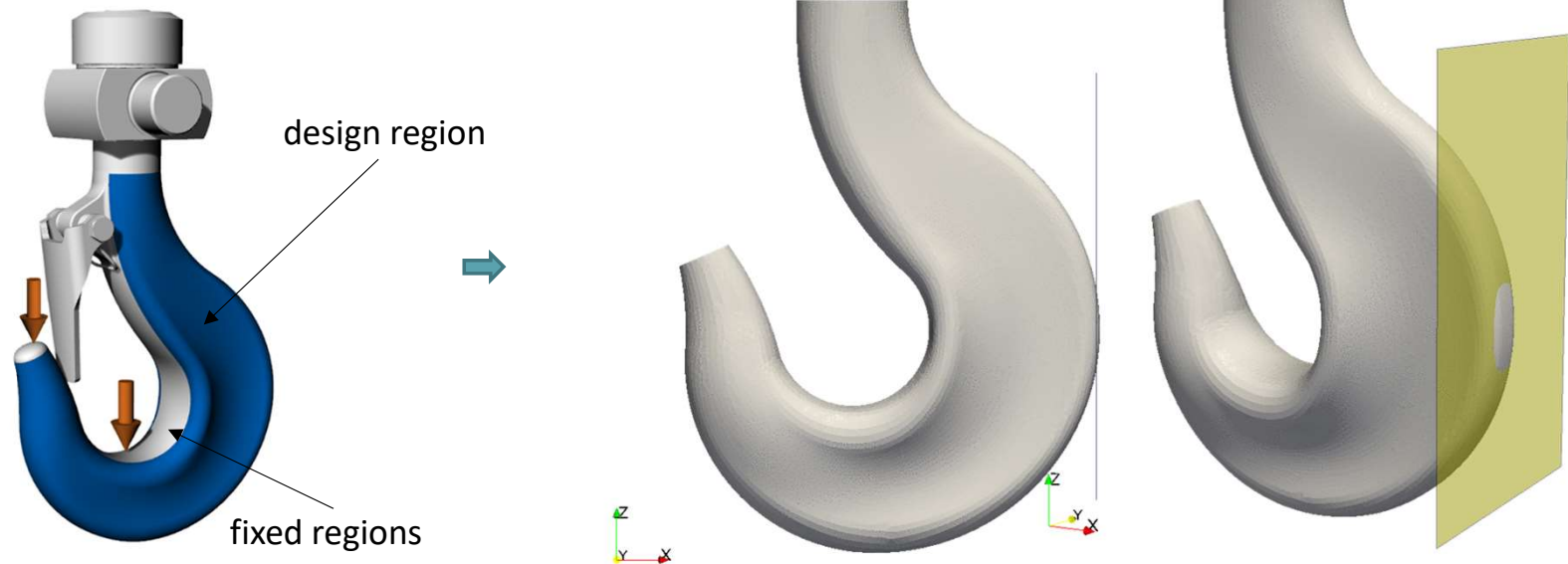
model [<https://grabcad.com/library/clutch-transmission>]

Courtesy of Armin Geiser



Optimization

- Stress adaptation with multi-objectives
 - minimize (mass + strain energy)
 - packaging violation (half space) ≤ 0



Courtesy of Armin Geiser





 <https://github.com/KratosMultiphysics>

 www.youtube.com/@kratosmultiphysics3578

